

Diwali Sales Data

Exploratory Data Analysis (EDA) Report

Prepared from: DIWALI_SALES_EDA.ipynb

Dataset: Diwali Sales Data.csv

1. Overview

This report documents an exploratory data analysis (EDA) performed on the Diwali Sales Data dataset. The analysis covers data cleaning, univariate distributions, and bivariate/multivariate relationships between customer demographics, geography, and purchase behavior.

1.1 Dataset Summary

Property	Value
Rows	11,251
Columns	15 (13 retained after cleaning)
Source file	Diwali Sales Data.csv
Key fields	Gender, Age, Marital_Status, State, Zone, Occupation, Product_Category, Orders, Amount

1.2 Column Data Types (raw)

Column	Non-Null Count	Dtype
User_ID	11,251	int64
Cust_name	11,251	object
Product_ID	11,251	object
Gender	11,251	object
Age Group	11,251	object
Age	11,251	int64
Marital_Status	11,251	int64
State	11,251	object
Zone	11,251	object
Occupation	11,251	object
Product_Category	11,251	object
Orders	11,251	int64
Amount	11,239	float64
Status	0	float64
unnamed1	0	float64

2. Data Cleaning

2.1 Dropped Columns

The columns 'unnamed1' and 'Status' were entirely empty (0 non-null values) and were dropped from the dataset:

- Code: `df = df.drop(columns=['unnamed1','Status'])`

2.2 Duplicate Records

A check for fully duplicated rows found 8 duplicate records.

2.3 Missing Values

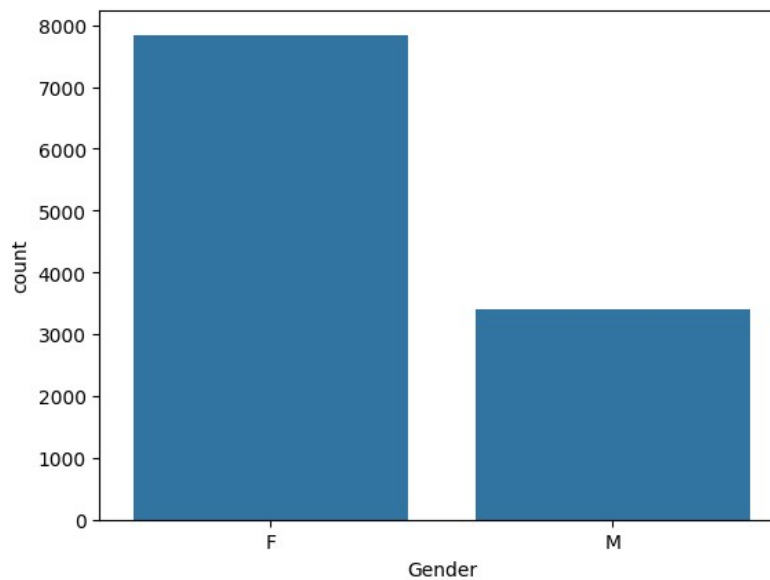
Before cleaning, the 'Amount' column had 12 missing values; all other columns were complete. Missing values in 'Amount' were imputed using the column median:

- Code: `df['Amount'] = df['Amount'].fillna(df['Amount'].median())`

After imputation, the dataset had zero missing values across all columns.

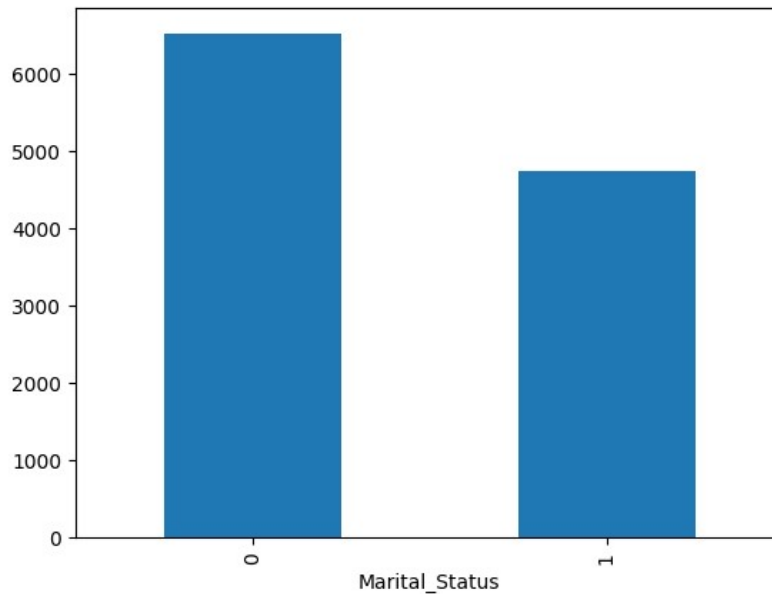
3. Univariate & Bivariate Analysis

3.1 Gender Distribution



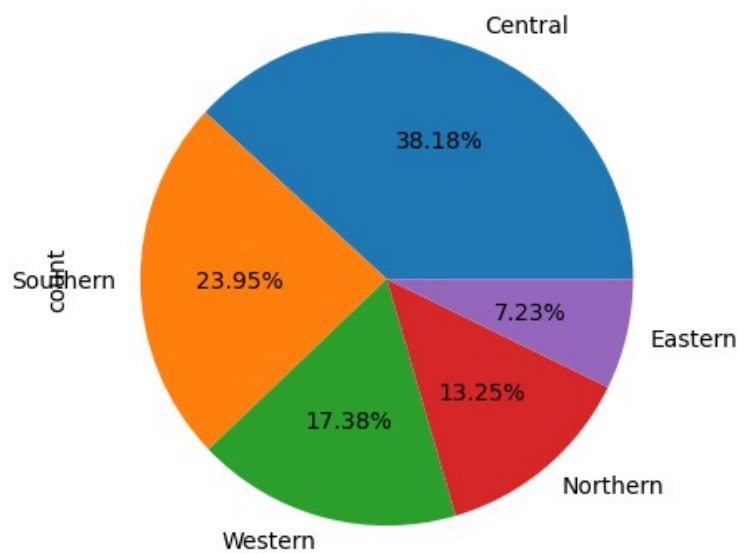
Count plot of customers by Gender. Female customers substantially outnumber male customers in the dataset, consistent with the finding that women purchase more goods than men in this sales period.

3.2 Marital Status Distribution



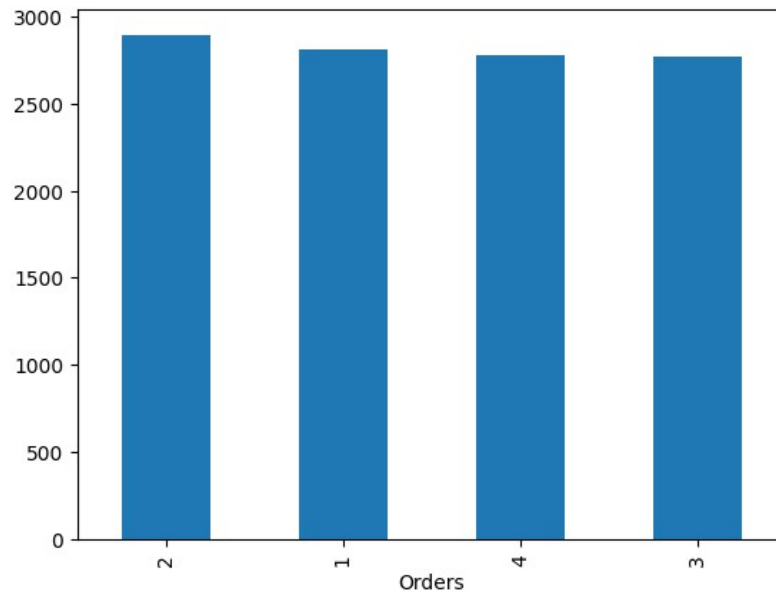
Bar chart of customer counts by Marital_Status (0 = unmarried, 1 = married). Unmarried customers make up the larger share of purchases.

3.3 Zone Distribution



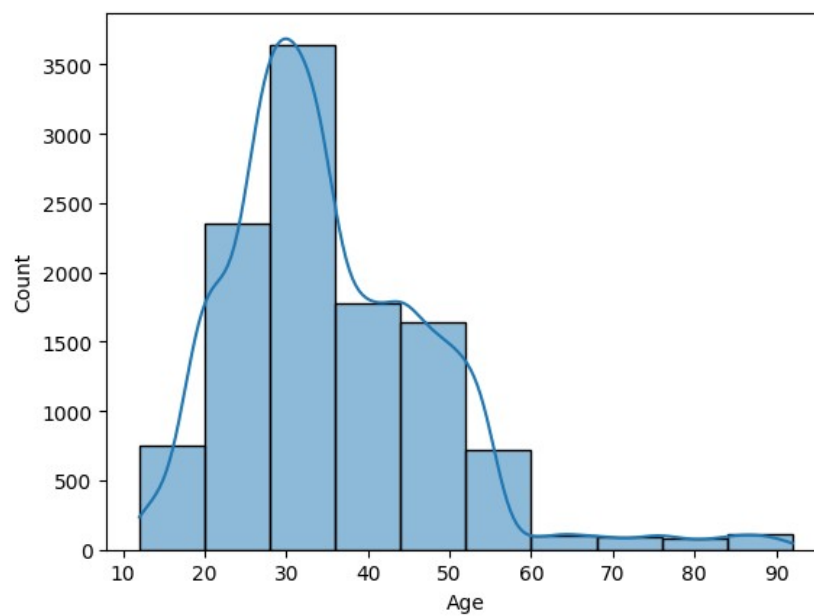
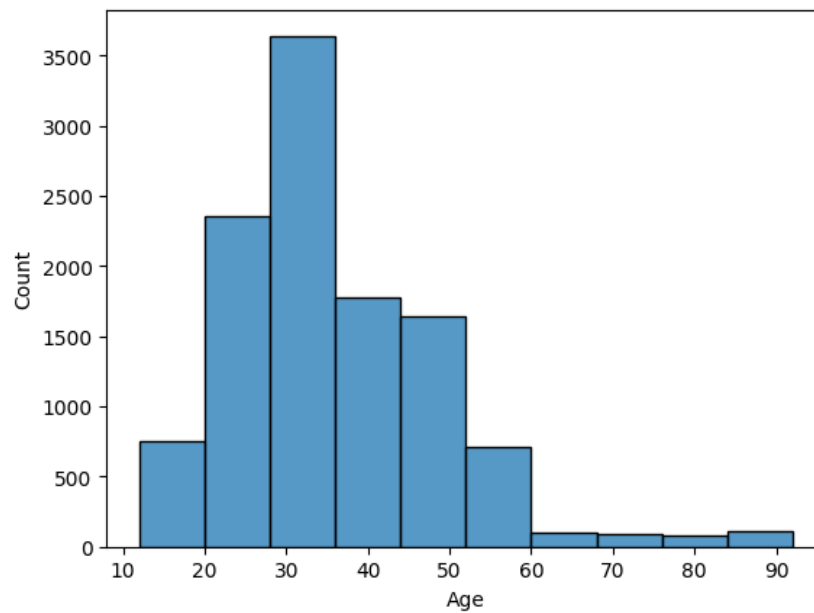
Pie chart showing the percentage share of transactions across zones. The Central zone accounts for the largest share of transactions, followed by Southern, Western, Northern, and Eastern.

3.4 Orders Distribution



Bar chart of order quantity frequency. An order quantity of 2 is the most common, indicating customers typically purchase in pairs rather than single units or bulk.

3.5 Age Distribution

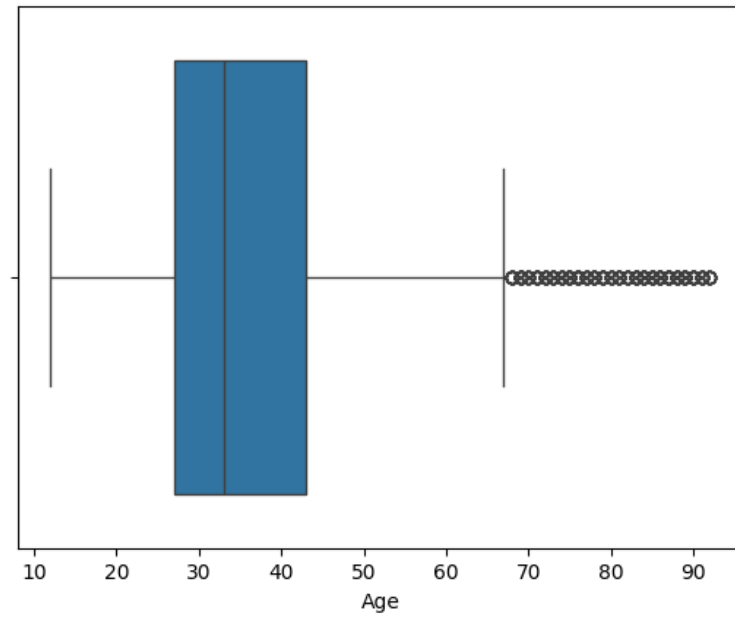


Age distribution with KDE overlay

Age skewness: 1.18 (positive / right-skewed).

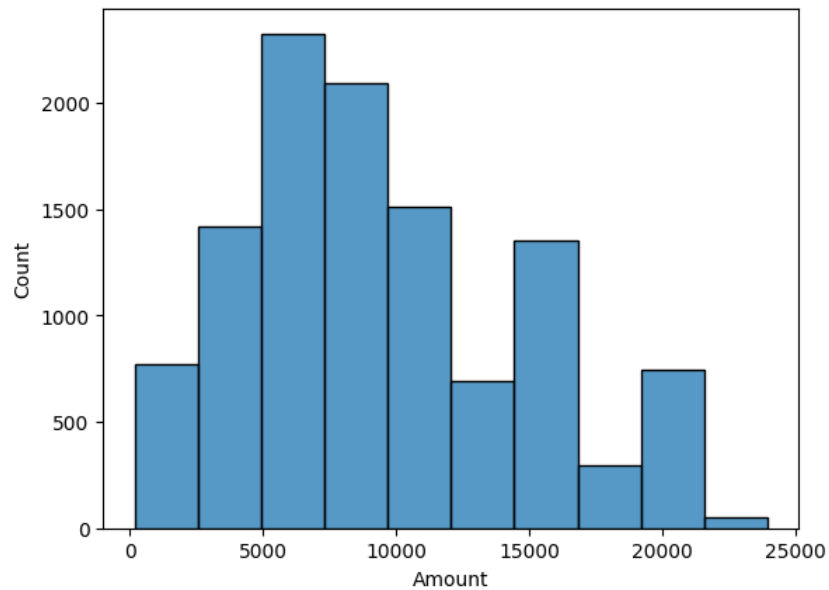
Most customers fall between approximately 20 and 50 years old, with the highest concentration around 25–35 years. Older customers above 60 are relatively rare, and this long right tail is the source of the positive skew.

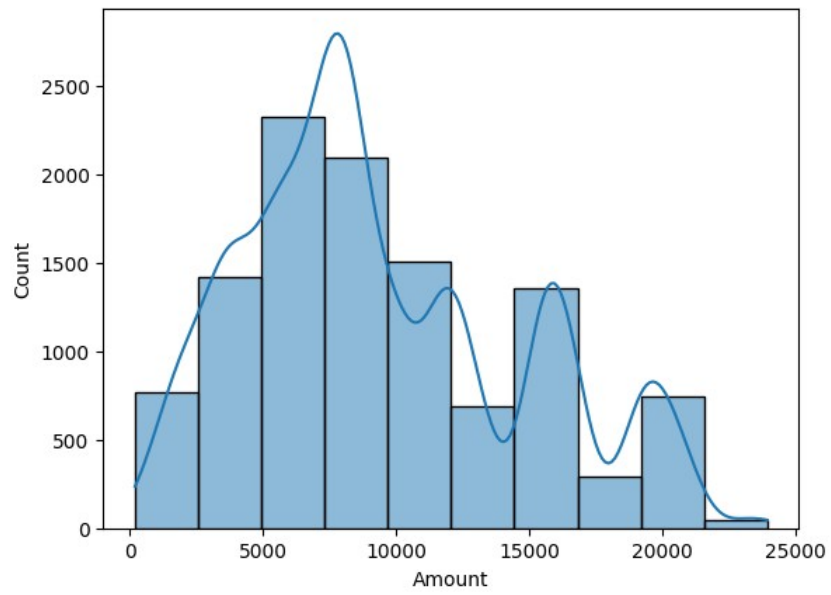
3.6 Age Boxplot



The boxplot shows most customers fall between roughly 27 and 43 years old, with a median age near 33. Several high-age observations above approximately 67 appear as outliers, reinforcing the right-skewed shape seen in the histogram.

3.7 Amount Distribution





Amount distribution with KDE overlay

Amount skewness: 0.56 (moderately right-skewed).

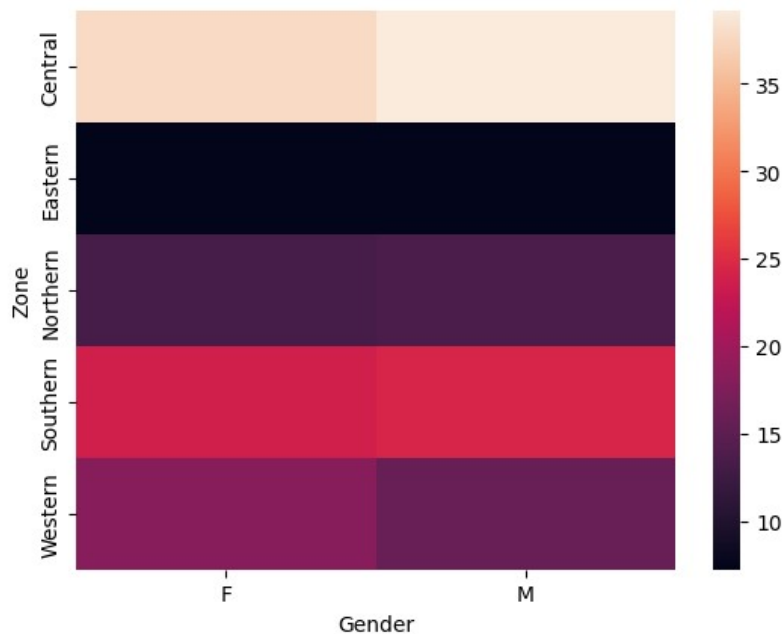
Most transactions are concentrated in the 5,000–10,000 range, with higher-value transactions becoming progressively less frequent; a small number of transactions extend beyond 20,000.

4. Multivariate Analysis

4.1 Zone vs. Gender

Cross-tabulation of transaction counts by Zone and Gender:

Zone	Female	Male
Central	2,961	1,335
Eastern	567	247
Northern	1,031	460
Southern	1,865	830
Western	1,418	537



Heatmap of Zone vs. Gender share (% normalized by gender column)

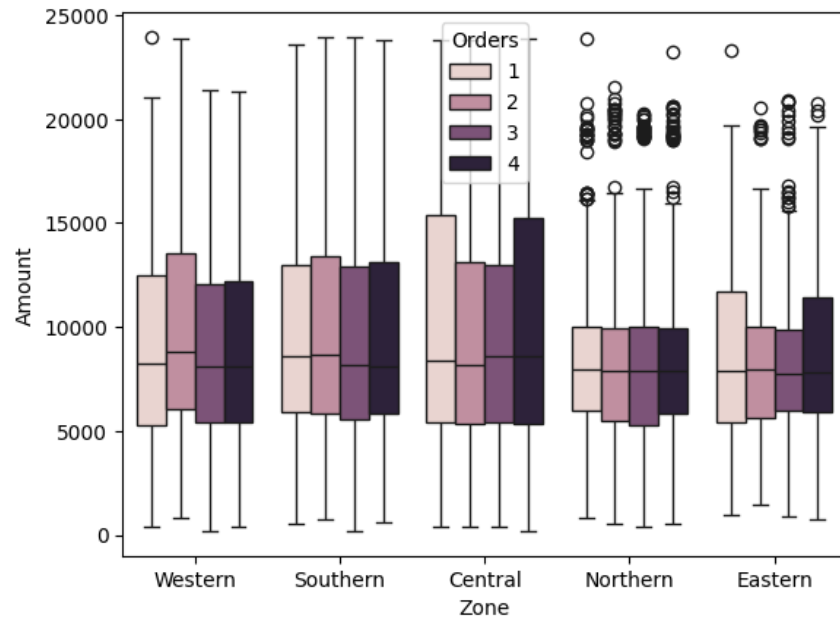
The Central Zone has the highest share of both female and male customers, while the Eastern Zone has the lowest share. The Southern Zone ranks second for both genders. Overall, the relative distribution of females and males across zones is similar, suggesting zone popularity is not strongly gender-dependent.

4.2 Top Product Categories by Sales

Rank	Product Category	Total Amount
1	Food	33,958,210.5
2	Clothing & Apparel	16,495,019.0
3	Electronics & Gadgets	15,643,846.0

Food is the top-selling product category by a wide margin, generating ~34M in total sales — more than double Clothing & Apparel (~16.5M) and Electronics & Gadgets (~15.6M), which are close to each other in second and third place.

4.3 Amount by Zone and Order Quantity



Boxplot of transaction Amount grouped by Zone and colored by number of Orders. Median order amounts are fairly similar across zones (roughly 7,500–10,000), regardless of order quantity. The Central zone shows the highest overall spread and median amounts, especially for single- and 4-order transactions. Northern and Eastern zones show many more high-value outliers (20,000+), indicating occasional large transactions well above the typical range. The Western zone has the tightest, most consistent distribution with the fewest extreme outliers.

5. Key Findings & Conclusion

- Female customers purchase more than male customers overall.
- Unmarried customers tend to make more purchases than married customers.
- The Central Zone has the highest sales activity compared to all other zones, for both genders.
- Customers most frequently order a quantity of 2 items per transaction.
- Customer age skews young-to-middle-aged, concentrated around 25–35 years, with a long tail of older outlier customers.
- Transaction amounts are right-skewed, typically falling between 5,000 and 10,000, with a minority of high-value outlier transactions.
- Food is by far the top-selling product category, generating over 2x the revenue of the next closest categories (Clothing & Apparel and Electronics & Gadgets).
- Order amount medians are broadly consistent across zones, but Central shows the highest spread, while Northern and Eastern have more high-value outliers and Western is the most consistent/least volatile.